

Shirley Achieng Ouma

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PROFESSIONAL SUMMARY

Detail-oriented software engineer with a strong technical mindset and excellent problem-solving skills. Self-motivated programmer with hands-on experience in application development, debugging, and collaborating on IT initiatives. Adaptable and fast learner with working knowledge of modern programming languages and a keen interest in emerging tech trends. Proven team player with cross-cultural communication skills and a track record of managing multiple responsibilities in dynamic environments.

AREAS OF EXPERTISE

Professional Skills: Customer service, Ability to analyze complex problems, identify solutions and implement efficiently, Effective communication skills and ability to work in a team, organizational skills, Research & Analysis skills and Adaptability & Initiative skills.

Technical Skills: **Network and Internet Security**

Collaboratively built a simple web app and secured it against common attacks.

Key Features: Implemented defenses against SQL injection, XSS, and CSRF attacks. Use HTTPS with TLS/SSL encryption.

Computer Network and Operating system

Collaboratively developed a python-based client-server file sharing application implementation of socket programming

Content Management Systems, Document Management Systems, Basic Database Skills, IT infrastructure Management Programming Languages (e.g., Python, Java, C++, JavaScript (ES6), Django, React, NodeJS, Rest API's, Cloud computing, ML, MySQL, MongoDB, Bootstrap, HTML-5, CSS-3, Git and GitHub, Docker, Visual Studio Code.

PROFESSIONAL EXPERIENCE

ICT Attaché, Kenya Civil Aviation Authority, Nairobi.

Aug 2023 – Oct 2023

KCAA ensures the highest level of safety in all civil aviation operations within Kenya's airspace by implementing hefty safety oversight programs, conducting inspections, audits, and

investigations to identify and address safety risks. I played a role in the planning, implementation, and support of the Authority's information and communication technology systems in the ICT department.

- ◆ **Network Design and Management:** Configured, and managed network infrastructure, ensuring secure and efficient data communication, website functionality and accessibility. Oversee the operation of communication systems, ensuring seamless communication and managing platforms and tools.
- ◆ **Cybersecurity:** Implemented cybersecurity measures, monitor breaches, and maintained security protocols especially during flight operators licensing registration Supported the implementation and configuration of firewalls, Intrusion Detection Systems (IDS), and Virtual Private Networks (VPNs) to secure critical aviation infrastructure.
- ◆ Participated in risk assessments and vulnerability scans, identifying potential security gaps in KCAA's IT systems and recommending mitigation strategies.
- ◆ Assisted in drafting and updating cybersecurity policies to ensure compliance with ICAO regulations and international aviation security standards.
- ◆ **System Maintenance and Support:** Provided technical support, troubleshooting, and maintaining IT systems to ensure reliability. Uploaded documents to the repository, edited KCAA website content, and also ensured smooth operation of the portal.
- ◆ **Data Analytics and Management:** Acquire abilities in collecting, analyzing, and managing data to support decision-making and maintain data integrity. Leveraged available data to create charts and presented findings via written reports. This is key for developing and conducting client satisfaction surveys and analyzing feedback.
- ◆ **Technical Documentation and Reporting:** Prepare technical documents, manuals, and reports, documenting system configurations and ICT activities.

SOFTWARE ENGINEER, Realm Elevators Company, Nairobi

Feb 2023 – Jul 2023

Realm elevators is a startup company that offers IT services.

- ◆ **Designed the realm elevators company's website,** we restructured the backend to be more efficient. Established and improved marketing plan for the company's social mediums, created training documents for department to train members on software and web implementations, collaborated with cross-functional teams to gather requirements and design software architecture, participated in stand-ups, sprint planning, and retrospectives to drive smooth project execution and lastly documented software defects with bug tracking system, reporting defects to software developers.
- ◆ **Created Progressive Web application (PWA)** that offers IT services and solutions such as web development, mobile app development etc. Tools: Django, JavaScript, HTML, CSS, Bootstrap 5, MySQL, SEO frameworks. <https://realmelevators.com/>
- ◆ Utilized Django Rest for the structure design of the backend and frontend UI library React and JS to **develop a simple task manager desktop application** as requested by potential client. The application is a sort of to-do list that enables user to add and save their daily tasks and also be able to edit and delete. The aim was to develop an app that helps people to manage their time by writing down what they need to do or the tasks they need to complete. Tools: React.JS, Django, MySQL.
- ◆ **Developed a restaurant website for a client,** that shows the menu of the restaurant. Customers are able to order food online and book reservation. The customer is able to login, choose what they have ordered by placing their order on cat, pay and book reservation. Tools: Django, JavaScript, HTML, CSS, Bootstrap 5, MySQL, AWS, SEO frameworks.
- ◆ **Created Bot Application like WhatsApp,** which is an automatic chat app that gives responses to users and mark read text and show unread text.

EDUCATION

THE UNIVERSITY OF NAIROBI

P.O. Box 30197 Nairobi, Kenya.

Bachelor of *Information/Computer Science*.

Sep 2020 - Dec 2024

St. Clare's Maragoli Girls High School

P.O Box 480-50300 Maragoli, Vihiga.

Jan 2016 - Nov 2019

REFEREES

Dr. Dorothy Njiraine Chair

DIS UON [Tel:0722649860](tel:0722649860)

The University of Nairobi

Zuwena Cheruon ICT

Analyst [Tel:0722502533](tel:0722502533)

Kenya Civil Aviation Authority

Nehemiah Wekesa

Senior developer at Realm

Elevators company

[Tel:0707967928](tel:0707967928)

Realm Elevators Company